Adaptive Momentum and Memorization Under Label Noise: A Controlled Study of Adam-NV

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Abstract

We investigate whether per-element adaptation of Adam's first-moment coefficient improves predictive accuracy and how it affects the fitting of corrupted training labels. Adam-NV decreases β_1 in response to a moment-derived score intended to reflect gradient noise. Across paired-seed comparisons on CIFAR-10 and CIFAR-100, ResNet-18 and a small vision transformer, we find no accuracy improvement that survives the stated multiplicity corrections. A candidate gain at learning rate 3×10^{-3} is $+0.17$ percentage points over ten seeds (95% unadjusted paired CI $[+0.05, +0.29]$; Holm-adjusted $p = 0.057$). Adam-NV incurs approximately 17% greater step time and twice the optimizer state in the reported setup. Its proposed noise score is not validated by direct diagnostics: a registered late-training analysis is unresolved because the direct estimate cannot reliably rank elements within tensors, while exploratory early-checkpoint analysis finds no within-tensor correlation. In contrast, under 20% symmetric label noise, Adam-NV fits 3.91 percentage points more corrupted labels than Adam on ResNet-18 (eight paired seeds, 95% CI $[+3.11, +4.71]$), without a detectable difference in held-out accuracy. Fixed-global-β_1, coefficient-shuffling, tensor-mean and tensor-subset controls reveal that tensor-wise coefficient structure explains a substantial portion of this behavior, but do not isolate the contribution of temporal adaptation. On the small vision transformer, a fixed matched coefficient yields a comparable memorization increase. In an additional five-seed, pilot-calibrated control, fixed per-tensor momentum coefficients preserve similar memorization trajectories: at epoch 60, Adam-NV fits 83.59% of corrupted labels versus 82.13% for the frozen control (frozen minus Adam-NV, -1.45 pp; unadjusted 95% paired CI $[-3.52, +0.62]$ pp). This supports a substantial role for a static tensor-level coefficient profile, without establishing equivalence or excluding additional effects of temporal adaptation. These findings characterize architecture-dependent changes in memorization rather than establish noise awareness or improved generalization.

1 Introduction

Adam uses a fixed first-moment decay coefficient β_1 even though the reliability of recent gradients can vary during training (Kingma & Ba, 2015). This motivates momentum coefficients that respond to gradient statistics. However, an adaptive coefficient introduces a fundamental identification problem: an observed improvement may follow from using a different average coefficient rather than from adapting it. Small evaluation sets and seed variation further complicate comparisons between optimizers (Bouthillier et al., 2021; Choi et al., 2019; Schmidt et al., 2021).

We study Adam-NV, which adjusts β_1 elementwise using a score derived from Adam's moment estimates. Our goal is to determine what its adaptation accomplishes, not to assume that the score estimates gradient noise. We distinguish two outcomes: predictive performance and the memorization of corrupted labels. The latter measures how thoroughly a model fits deliberately corrupted training targets; it is not a direct measure of generalization harm.

The experiments progress from accuracy benchmarks and direct proxy diagnostics to label-noise interventions, including matched constant coefficients, within-tensor coefficient shuffling, tensor-mean substitution,

and restrictions to tensor subsets. We use paired seeds and report effect sizes and uncertainty. Some runs were motivated by prior exploratory observations; we document their chronology and do not treat every resulting comparison as independently confirmatory.

Contributions.

1. **A controlled accuracy evaluation.** Across the tested CIFAR-scale settings, Adam-NV has no detected accuracy gain after the stated multiplicity adjustments. We document a small candidate effect at one learning rate, the limits of its inference, and measured computational overhead.
2. **A distinct memorization effect.** On ResNet-18 with 20% symmetric label noise, Adam-NV memorizes 3.91 percentage points more corrupted labels than Adam (eight paired seeds, 95% CI $[+3.11, +4.71]$), with no detected difference in held-out accuracy. The effect is also observed in the tested additional noise rates and small ViT setting.
3. **Mechanistic controls with qualified interpretation.** Global constant-β_1 controls do not reproduce the full ResNet-18 result in the tested range, whereas tensor-mean and within-tensor shuffle controls preserve a substantial portion. These interventions implicate tensor-level coefficient structure. A further independent-pilot control with frozen per-tensor coefficients retains similar memorization trajectories over five paired seeds, although its interval is too broad to establish equivalence with Adam-NV. A fixed matched global coefficient produces a comparable increase on the ViT.
4. **A direct check of the noise-proxy rationale.** The registered late-training diagnostic is unresolved; exploratory early-checkpoint measurements do not support a within-tensor relationship to estimated noise. The score also has a gradient-scale dependence, limiting its interpretation as a noise estimator.

2 Related Work

Adaptive optimizers. Adam (Kingma & Ba, 2015) combines momentum with per-element step sizes from a running second moment. Later variants repair its convergence behavior (AMSGrad (Reddi et al., 2018)), stabilize early steps (RAdam (Liu et al., 2020)), add Nesterov momentum (Nadam (Dozat, 2016)), decouple weight decay (AdamW (Loshchilov & Hutter, 2019)), or scale steps by the deviation of the gradient from its running mean (AdaBelief (Zhuang et al., 2020)). All keep β_1 fixed. Classical momentum (Polyak, 1964; Sutskever et al., 2013) is likewise used with a constant coefficient.

Adapting momentum. GONO (Gera, 2026) sets β_1 per parameter tensor from the cosine similarity of consecutive gradients, so that directionally consistent gradients receive more momentum. It is the closest published β_1-adaptive method to Adam-NV and is therefore our main adaptive baseline. The gradient noise scale (McCandlish et al., 2018) relates noise to the useful batch size, which motivates using a noise estimate to steer optimizer hyperparameters.

Label noise and memorization. Deep networks fit random labels (Zhang et al., 2017), learn clean patterns before corrupted ones (Arpit et al., 2017), and early stopping helps under label noise (Li et al., 2020). We use label corruption as a controlled setting in which the adaptive part of an optimizer has a chance to matter, and we measure memorization directly.

Evaluating optimizers. Optimizer comparisons depend strongly on tuning and on seed variance (Wilson et al., 2017; Choi et al., 2019; Schmidt et al., 2021; Bouthillier et al., 2021). We follow this line by pairing runs by seed, reporting uncertainty, and adding controls for the quantity that the proposed method adapts.

3 Methods

3.1 Adam-NV

All operations are elementwise. Let g_t be the gradient at step t, m_t and v_t the first and second moment estimates, and $\beta_{1,t}$ the momentum coefficient. Adam-NV computes a moment-based score, designed as a

noise proxy, from the moments of the previous step,

$$s_t = \frac{\max\big(v_{t-1} - m_{t-1}^2,\, 0\big)}{v_{t-1} + \lambda\,|m_{t-1}| + \epsilon}, \tag{1}$$

smooths it with a warm-up weight and an exponential average,

$$\rho_t = \alpha\,\rho_{t-1} + (1-\alpha)\,\frac{t}{t+k}\,s_t, \tag{2}$$

and maps it to a momentum coefficient that can only fall below the base value β_{base}:

$$\beta_{1,t} = \beta_{\min} + (\beta_{\text{base}} - \beta_{\min})\max(1 - \gamma\rho_t,\, 0)\,. \tag{3}$$

The moment updates use the time-varying coefficient,

$$m_t = \beta_{1,t}\,m_{t-1} + (1-\beta_{1,t})\,g_t, \tag{4}$$

$$v_t = \beta_2\,v_{t-1} + (1-\beta_2)\,g_t^2. \tag{5}$$

Because $\beta_{1,t}$ varies, the usual factor $1-\beta_1^t$ is replaced by $1-P_t$, where $P_t = \max\big(\prod_{s\le t}\beta_{1,s},\, 10^{-7}\big)$, and the update is

$$\theta_{t+1} = \theta_t - \eta\,\frac{m_t/(1-P_t)}{\sqrt{v_t/(1-\beta_2^t)} + \epsilon}. \tag{6}$$

Starting from $m_0 = 0$, the weights that m_t puts on past gradients sum to $1 - \prod_{s\le t}\beta_{1,s}$ (by induction from (4)), so dividing by $1-P_t$ normalizes them exactly. State is initialized to zero ($m_0 = v_0 = \rho_0 = 0$, $P_0 = 1$). The implementation supports parameter groups and L2 weight decay (default 0), skips parameters without gradients, and does not support sparse gradients, AMSGrad or fused kernels. Unless stated otherwise we use $\beta_{\text{base}} = 0.9$, $\beta_{\min} = 0.1$, $\beta_2 = 0.999$, $\gamma = 1$, $\alpha = 0.95$, $k = 100$, $\lambda = 0.1$ and $\epsilon = 10^{-8}$, without tuning.

The numerator estimates a second central moment with exponential averages and the ratio is bounded by one. The proxy is not a calibrated variance: because v has a much longer memory than m, it also rises when gradient magnitudes shrink during training, and the term $\lambda|m|$ has the units of a gradient where v has those of its square, so the score depends on the absolute gradient scale and not only on a noise fraction. We did not validate it against a direct measurement of gradient noise when we designed the method; a diagnostic that also measures this scale dependence is reported in Section 5.6.

With $\gamma = 0$, the same ϵ, no weight decay and zero-initialized state, the method equals Adam apart from the clamp on P_t, which is reached after about 150 steps for $\beta_1 = 0.9$ and changes the bias-correction factor by at most 10^{-7}.

The response in (3) is the *distrust* response: a high score lowers β_1. The *smooth* response is an ablation, not a proposed alternative: it reverses the direction, so that a high score raises β_1 and noisy gradients are averaged over a longer window:

$$\beta_{1,t} = \beta_{\text{base}} + (\beta_{\max} - \beta_{\text{base}})\min(\gamma\rho_t,\, 1)\,, \tag{7}$$

with $\beta_{\max} = 0.999$ and the other quantities as above, so that $\beta_{1,t} \in [0.9,\, 0.999]$.

Two controls act on the per-element coefficient $\beta_{1,t}^{(j,i)}$ of element i in parameter tensor j with n_j elements, after (3) and before (4). The *shuffle* control replaces it by $\beta_{1,t}^{(j,\pi_t^{(j)}(i))}$, where $\pi_t^{(j)}$ is a uniformly random permutation of the n_j elements, drawn afresh at every step or once per tensor. The *tensor-mean* control replaces it, for every element of tensor j, by $\bar\beta_{1,t}^{(j)} = n_j^{-1}\sum_i \beta_{1,t}^{(j,i)}$. In both, ρ_t is still computed from each element's own gradients and P_t is formed from the coefficients actually applied. Two further variants lower the second-moment decay β_2 or give the score its own decay (*decoupled*); they are used only in the non-stationary noise experiments and are compared with Adam-NV itself.

3.2 Baselines and controls

Adam uses $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and no weight decay. **AdamW** uses weight decay 10^{-4}. **AdaBelief** uses the standard form without rectification. **GONO** sets β_1 per tensor from the cosine similarity c_t of consecutive gradients, $\beta_1 = \text{clip}(\beta_{\text{base}}(1 + \lambda_G c_t), \beta_{\text{lo}}, \beta_{\text{hi}})$ with $\lambda_G = 0.4$, $\beta_{\text{lo}} = 0.5$ and $\beta_{\text{hi}} = 0.99$. We re-implemented GONO following the paper, with the per-tensor cosine and cosine $\epsilon = 10^{-10}$ of the authors' released code. Our ϵ placement in the Adam step follows the paper and PyTorch, whereas the released code places it differently; on synthetic gradients the two implementations differ by at most 2.5×10^{-6} in parameter value, and they agree to 10^{-15} once both differences are matched.

The key control is **constant-β_1 Adam**, the Adam-NV code with $\gamma = 0$ and β_1 fixed at a chosen value. In every setting we first take the time average of Adam-NV's logged β_1 and set the control to that value. We used the logged statistic because it was the only summary of β_1 available when the controls were designed. It is a mean over parameter tensors that gives every tensor the same weight; the element-weighted mean is higher (Section 5.7), so the matched constants are lower than the controller's element-level average, and the range of constants we test (0.70 to 0.95) brackets both. Agreement with torch Adam for a fixed β_1 is checked by unit tests (float64, 500 steps, tolerance 10^{-6}).

Subset controls. To enable the controller on only some tensors, the optimizer holds two parameter groups that differ only in γ (1 for the selected tensors, 0 for the rest), so the rest keep $\beta_1 = \beta_{\text{base}}$ (0.9 up to a float32 rounding of 4×10^{-8}) and Adam's update; tensors are split by rank (at most one: 41 tensors; at least two: 21). A constant-β_1 variant gives the selected group $\gamma = 0$ and $\beta_1 = c$ instead, with plain Adam on the rest. In a gradient-replay test on ResNet-18 tensor shapes (float32, 220 steps) the plain tensors differed from torch Adam by at most 3×10^{-7} and the adaptive tensors from a stand-alone Adam-NV by zero. The plain group still runs through the Adam-NV code, so it is not bit-identical to torch Adam, and we ran the whole model with $\gamma = 0$ and $\beta_1 = 0.9$ through that code as a null control (eight seeds, 20% label noise, rule fixed beforehand). It fits 0.15 pp fewer corrupted labels than Adam (CI $[-0.88, +0.58]$, higher in 5 of 8 seeds); the interval is wider than the ± 0.5 pp equivalence margin fixed beforehand, so the outcome is inconclusive (a margin of ± 1 pp, which was not registered, would have been met). The paired standard deviation of this difference, 0.88 pp, is the trajectory-level noise floor of the code path, close to that of Adam-NV minus Adam (0.95 pp).

Pilot-calibrated frozen tensor-wise control (Experiment 22). We added a control that assigns each of ResNet-18's 62 parameter tensors a separate but fixed first-moment coefficient. A distinct Adam-NV pilot (seed 100, disjoint from the evaluation seeds 0–4) supplied time-averaged per-tensor coefficients sampled during training; the resulting coefficient map, ranging from 0.625 to 0.898, was frozen before evaluation and reused unchanged in all five evaluation seeds. Adam-NV and the frozen control were paired within evaluation seed, with otherwise matching training configurations (CIFAR-10, 20% symmetric label corruption, ResNet-18, 60 epochs, learning rate 10^{-3} and batch size 128). This design removes temporal and within-tensor coefficient variation simultaneously while retaining pilot-derived differences between tensors. The pilot estimates a descriptive coefficient profile; it does not independently identify gradient noise. Our exploratory analysis reports two-sided paired t intervals without multiplicity adjustment and does not claim equivalence.

Checkpoint trajectories (Experiment 23). The same runs provide epoch-wise held-out test accuracy and corrupted-label memorization evaluated at epochs 1, 5, 10, 20, 30, 45 and 60. The memorization metric is the fraction of intentionally corrupted training examples predicted as their assigned wrong labels. Checkpoint values are descriptive repeated measurements from five runs, not independent replicates across epochs. Unlike several earlier experiments that use a last-five-epoch mean, the new memorization comparisons are at the fixed *epoch-60 checkpoint*; therefore we do not numerically pool their estimates with those earlier results.

4 Experimental Setup

Data and model. We use CIFAR-10 and CIFAR-100 (Krizhevsky, 2009) (50,000 training and 10,000 test images) with a ResNet-18 (He et al., 2016) whose stem is adapted to 32×32 inputs. Training uses random crop with padding 4 and horizontal flip, cross-entropy loss, batch size 128, 60 epochs ($\approx$23,400 steps) and cosine-annealed learning rate (Loshchilov & Hutter, 2017). Learning rate is 10^{-3} unless stated. Initialization,

batch order, gradient noise and label corruption each use dedicated random generators seeded from the run seed (seed, seed, seed+1 and seed+2000), so within a seed every optimizer sees the same initial weights, batches, noise directions and corrupted labels. The CIFAR test set serves as the validation set and every result is the value at fixed epochs, with no checkpoint selection. Software was PyTorch (Paszke et al., 2019) 2.10.0 (CUDA 12.8) with Python 3.12.13 on Tesla T4 GPUs, with deterministic mode enabled.

Experiments. Table 1 lists every experiment. Gradient noise, when injected, is norm-scaled Gaussian noise (Neelakantan et al., 2015): for every parameter tensor, a random direction of unit Frobenius norm scaled by $\sigma\|g\|_F$ is added to its gradient, so with $\sigma = 2.0$ the perturbation is twice as large as the gradient. It is applied during the middle third of training (epochs 21–40, *phased*), ramped linearly from zero to σ over training (*ramp-up*), or in alternating five-epoch noisy and clean blocks (*intermittent*).

For label noise, 20% of the training labels are replaced once by a random different class; the corrupted subset is determined by the run seed, so every optimizer sees the same corruption within a seed, and the test labels are untouched.

At the end of every epoch we pass the training set through the network in evaluation mode and record the fraction of corrupted examples predicted as their *wrong* label (the memorization rate) and the accuracy on uncorrupted examples; we average the last five epochs to reduce epoch-to-epoch fluctuation. This measures how strongly corrupted labels are fitted; it is not a measure of generalization.

Metrics and statistics. The primary accuracy metric is the mean validation accuracy over the last five epochs; we also report epochs to 90% accuracy and the validation-loss area under the curve (sum over 60 epochs). Runs are paired by seed. For each contrast we report the mean paired difference in percentage points (pp), a t-based 95% confidence interval, the paired t-test p-value, the exact two-sided sign-flip p-value (with $n = 4$ it cannot fall below 0.125), and Holm-adjusted p-values (Holm, 1979) within each experiment's family of contrasts. All differences are computed from unrounded per-seed values, so they can differ by 0.01 pp from differences of the rounded means shown in the tables. Confidence intervals are unadjusted 95% t intervals; only the p-values labelled Holm are adjusted for multiplicity. Inference rests on the paired t interval with Holm adjustment; the exact sign-flip test uses the same paired differences without a normality assumption and serves as a check, and the two can disagree in small samples (with four seeds the sign-flip p is at least 0.125 even when all four differences have the same sign). We use three labels: a *detected* difference (Holm-adjusted $p < 0.05$), *no detectable* difference (the interval includes zero), and *practical equivalence*, which we claim only if the whole 95% interval lies within ± 0.15 pp. No comparison in this paper currently meets the equivalence bar, so "no detectable difference" never means "equal".

Chronology and test-set use. The clean benchmark and the learning-rate experiment came first and were exploratory. The learning-rate gain motivated the constant-β_1 control, and a memorization difference seen in a label-noise diagnostic motivated the label-noise controls; these follow-up experiments used decision rules written in run books before their runs. No hyperparameter, stopping or checkpoint choice used the test set, but later experiments were chosen after earlier test-set results were seen, so p-values that pool exploratory seeds are optimistic. The ten-seed extension at learning rate 3×10^{-3} adds six new seeds (4–9) whose results are also reported separately from the exploratory seeds 0–3. The tensor-subset experiment was designed after Table 8 had been seen and used a rule written beforehand; its additivity and multiplicity analyses are exploratory. Most logs record library versions, arguments and SHA-256 hashes of the source files; the earliest label-noise logs and the first GONO logs do not, or carry older hashes, so we repeated two label-noise runs and one GONO run with the final code, and each reproduced the earlier log exactly.

5 Results

The results follow the two purposes of the study. The first subsections evaluate accuracy, which gives a controlled negative result; they are followed by the computational cost and a test of whether the score measures gradient noise. The last two subsections study memorization under label noise, which gives our strongest findings, on ResNet-18 and on a second architecture.

Table 1: Overview of experiments. All use ResNet-18, 60 epochs and batch size 128 (timing excepted).

Experiment	Setting	Arms	Seeds	Decision rule fixed beforehand
Clean benchmark	CIFAR-10, lr 10^{-3}	Adam, Adam-NV, AdamW, GONO (and a GONO variant), AdaBelief	10	no (descriptive)
Learning-rate sensitivity	CIFAR-10, lr 3×10^{-3} and 3×10^{-4}	Adam, Adam-NV, GONO	4	no (exploratory)
Constant-β_1 control	CIFAR-10, lr 3×10^{-3}	Adam, Adam-NV, Adam with $\beta_1 = 0.83$ and 0.80	10 (4 exploratory, 6 new)	yes
Non-stationary gradient noise	phased, ramp-up, intermittent; $\sigma = 2.0$	Adam, Adam-NV and β_2 variants	4 / 4 / 8	no (Holm-adjusted contrasts)
Controller direction, CIFAR-100	distrust vs smooth response; CIFAR-100 clean and intermittent	Adam-NV modes; Adam vs Adam-NV	4; 8 / 4	no (Holm-adjusted contrasts)
Label noise, control	20% label noise, lr 10^{-3}	Adam, Adam-NV, constant $\beta_1 = 0.81$	8	yes
Label noise, dose-response	same	constant $\beta_1 = 0.78$ and 0.70	8	yes
Label noise, response direction	same	Adam-NV with smooth response	8	yes
Label noise, constants above 0.9	same	constant $\beta_1 = 0.92$ and 0.95	8	yes
Label noise, β_1 shuffle	same	Adam-NV with β_1 permuted (fresh each step, or one fixed permutation); no-shuffle canary	8 (canary 2)	yes
Label noise, tensor mean	same	Adam-NV with β_1 set to the tensor mean; unchanged Adam-NV for per-tensor statistics	8 (canary 2)	yes
Label noise, other rates	10% and 40%, lr 10^{-3}	Adam, Adam-NV	6	yes
Label noise, null run	same	Adam-NV code with $\gamma = 0$, $\beta_1 = 0.9$, against Adam	8	yes
Label noise, constant on 1-D tensors	same	constant $\beta_1 = 0.79$ on the 41 tensors of rank ≤ 1, plain Adam elsewhere	8	yes
Label noise, tensor subsets	same	Adam-NV with the controller only on the 41 tensors of rank ≤ 1, or only on the 21 of rank ≥ 2	8	yes
Noise-proxy diagnostic	clean and 20% label noise, lr 10^{-3}	Adam-NV; direct gradient-noise measurement at 7 checkpoints ($K = 16$, then $K = 64$)	2, then 5	yes
Second architecture	ViT-tiny, lr 5×10^{-4}, clean and 20% label noise	Adam, Adam-NV, constant $\beta_1 = 0.85$ (clean and noise); constants 0.70, 0.80, 0.92, 0.95 (noise)	5	yes
Optimizer overhead	Tesla T4, random batch of 128	Adam, Adam-NV, AdamW, AdaBelief	20 rounds	not applicable
Frozen tensor-wise momentum and trajectories (Exps. 22–23)	CIFAR-10, 20% label noise, lr 10^{-3}	Adam-NV; pilot-calibrated static per-tensor β_1	1 pilot (seed 100); 5 evaluation pairs (0–4)	exploratory paired contrasts

5.1 Clean benchmark

Table 2 shows that Adam-NV, Adam and AdamW are statistically indistinguishable on clean CIFAR-10. Adam-NV exceeds Adam by 0.06 pp (95% CI $[-0.05, +0.16]$, Holm $p = 0.29$), a smaller gap than the +0.12 pp seen with the first four seeds, which is what seed noise would predict. AdamW's gap is similar (+0.08 pp). GONO and AdaBelief are both clearly lower than Adam (-0.70 pp, CI $[-0.82, -0.59]$, and -0.71 pp, CI $[-0.80, -0.62]$) and than Adam-NV (+0.76 pp, CI $[+0.62, +0.90]$, and +0.77 pp, CI $[+0.63, +0.90]$; 10 of 10 seeds each; all Holm $p < 10^{-5}$). Holm adjustment covers the six contrasts of this experiment (Adam-NV, AdamW, GONO and AdaBelief against Adam, and Adam-NV against GONO and AdaBelief). A GONO variant that uses Adam-NV's exact running-product bias correction instead of the paper's base correction gives $92.38 \pm 0.12\%$, within 0.06 pp of the base version ($p = 0.27$) and still 0.76 pp below Adam, so GONO's deficit is not caused by its bias-correction choice.

Table 2: Clean CIFAR-10, learning rate 10^{-3} (mean ± SD over seeds).

Optimizer	Seeds	Acc. (%)	Epochs to 90%	Val.-loss AUC
Adam	10	93.15 ± 0.11	19.0	26.31
Adam-NV	10	93.20 ± 0.15	18.6	26.39
AdamW	10	93.22 ± 0.15	18.4	26.43
GONO	10	92.44 ± 0.14	25.4	30.39
AdaBelief	10	92.44 ± 0.17	24.3	26.67

One GONO log lost its final row; the run was repeated and reproduced the other 59 epochs exactly.

5.2 Learning-rate sensitivity and the constant-β_1 control

Table 4 shows the only candidate accuracy gain, found in this exploratory four-seed experiment. At learning rate 3×10^{-3} Adam-NV exceeds Adam by +0.20 pp (95% CI $[+0.11, +0.28]$, higher in 4 of 4 seeds, Holm $p = 0.014$ within the four learning-rate contrasts); at 3×10^{-4} the difference is -0.02 pp. GONO is 1.54 pp below Adam at 3×10^{-3} and reaches 90% accuracy about 16 epochs later, and ties Adam at 3×10^{-4}. Adam-NV's logged β_1 averaged 0.83 at 3×10^{-3} and never exceeded its base value 0.9; the lowest value was 0.57. These logged statistics are means over the elements of each parameter tensor, then over tensors, so the minimum is the lowest tensor mean and not the lowest element.

Constant-β_1 control and seed extension. The four-seed controls in Table 4 gained +0.17 and +0.20 pp over Adam, as much as Adam-NV. Those four seeds, however, generated the hypothesis, and Adam happened to score low on them (92.94% against 93.07% on seeds 4–9). We therefore ran seeds 4–9 for all four arms under a rule fixed beforehand (Table 3). Over ten seeds Adam-NV exceeds Adam by +0.17 pp (95% CI $[+0.05, +0.29]$, higher in 9 of 10 seeds, raw $p = 0.011$, Holm $p = 0.057$ over the five contrasts of this experiment). Its gain is similar on the exploratory seeds (+0.20 pp) and on the new seeds alone (+0.16 pp, CI $[-0.08, +0.39]$, 5 of 6 seeds). The constant controls do not hold up: they gain +0.06 and +0.07 pp over Adam on ten seeds and -0.02 and -0.03 pp on the new seeds alone. Adam-NV minus the controls is +0.116 pp (CI $[-0.021, +0.253]$) and +0.105 pp (CI $[-0.035, +0.246]$), with Adam-NV higher in 8 of 10 seeds in both cases (Holm $p = 0.35$ and 0.37). By the pre-specified rule the outcome is inconclusive: the controls reproduce less than half of the gain, but the interval of the Adam-NV-minus-control difference includes zero. The data are compatible with an Adam-NV advantage of up to about 0.25 pp over a constant β_1, and with none; with the observed paired spread (SD 0.19 pp), about 31 paired seeds would be needed to detect a difference of 0.10 pp (80% power, two-sided $\alpha = 0.05$, no multiplicity correction), and 24 for the observed 0.116 pp. The final-epoch analysis gives the same picture (Adam-NV minus Adam +0.16 pp, Holm $p = 0.13$).

5.3 Non-stationary gradient noise

Table 5 shows that none of the three schedules produces a significant difference between Adam-NV and Adam, and the largest gap favors Adam in the intermittent schedule (-0.16 pp, CI $[-0.41, +0.09]$). Two follow-up families vary Adam-NV's second-moment decay and are compared with Adam-NV itself, not with Adam. In the first, a lower decay $\beta_2 \in \{0.99, 0.95\}$ is compared with the default 0.999 for each of four schedules (eight

Table 3: Ten-seed confirmation at learning rate 3×10^{-3} (last-five accuracy; seeds 0–3 exploratory, seeds 4–9 new). Holm over the five contrasts.

Arm	Seeds	Acc. (%)	Δ (pp)	95% CI	Holm p	Seeds above
Adam	10	93.02 ± 0.16	—	—	—	—
Adam-NV	10	93.19 ± 0.16	+0.17	$[+0.05, +0.29]$	0.057	9/10
const. $\beta_1 = 0.83$	10	93.07 ± 0.21	+0.06	$[-0.11, +0.22]$	0.82	7/10
const. $\beta_1 = 0.80$	10	93.08 ± 0.18	+0.07	$[-0.11, +0.24]$	0.82	8/10
Adam-NV − const. 0.83	10	—	+0.12	$[-0.02, +0.25]$	0.35	8/10
Adam-NV − const. 0.80	10	—	+0.11	$[-0.04, +0.25]$	0.37	8/10

Table 4: Learning-rate sensitivity and constant-β_1 controls (four seeds, last-five accuracy, mean ± SD).

LR	Optimizer	Acc. (%)	Δ vs Adam (pp)	95% CI	Holm p
3×10^{-3}	Adam	92.94 ± 0.11	—	—	—
	Adam-NV	93.13 ± 0.14	+0.20	$[+0.11, +0.28]$	0.014
	GONO	91.40 ± 0.25	−1.54	$[-2.03, -1.05]$	0.009
	const. $\beta_1 = 0.83$	93.11 ± 0.15	+0.17	$[-0.03, +0.36]$	0.28
	const. $\beta_1 = 0.80$	93.14 ± 0.16	+0.20	$[-0.11, +0.52]$	0.39
3×10^{-4}	Adam	92.33 ± 0.08	—	—	—
	Adam-NV	92.30 ± 0.16	−0.02	$[-0.31, +0.26]$	1.00
	GONO	92.18 ± 0.31	−0.15	$[-0.77, +0.47]$	1.00

Holm families: the learning-rate experiment (Adam-NV and GONO against Adam at each learning rate, four contrasts) and the control experiment (each control against Adam, and Adam-NV against each control, four contrasts) are adjusted separately.

contrasts, four seeds each). One contrast is Holm-significant: under ramped noise, $\beta_2 = 0.95$ is 0.16 pp *below* the default (CI $[-0.22, -0.10]$, Holm $p = 0.028$; the exact sign-flip p is 0.125, its minimum for four seeds). The largest gain is +0.44 pp for $\beta_2 = 0.95$ under intermittent noise (CI $[+0.06, +0.82]$, raw $p = 0.035$, Holm $p = 0.25$), and the remaining six contrasts have Holm $p \geq 0.64$. In the second family, a separate decay for the score is compared with the coupled value (eight contrasts); the smallest Holm p is 0.068 (ramped noise, decay 0.99, +0.13 pp).

Table 5: Adam-NV minus Adam under injected gradient noise (last-five accuracy).

Schedule	Seeds	Adam-NV	Adam	Δ (pp) [95% CI]	Holm p
Phased	4	92.86	92.75	+0.11 $[-0.35, +0.56]$	1.00
Ramp-up	4	92.97	92.95	+0.02 $[-0.22, +0.26]$	1.00
Intermittent	8	92.37	92.53	−0.16 $[-0.41, +0.09]$	0.51

5.4 Controller direction and CIFAR-100

Letting β_1 also rise above 0.9 (smooth response) does not help: the default distrust response is nominally ahead by +0.10 pp (clean), +0.09 pp (constant noise) and +0.29 pp (intermittent noise; CI $[+0.01, +0.57]$), but no Holm-adjusted value is below 0.14 (Table 6). On CIFAR-100 Adam-NV is 0.29 pp below Adam on clean data (eight seeds, CI $[-0.62, +0.04]$, Holm $p = 0.15$, Adam higher in six of eight seeds) and tied under intermittent noise (−0.01 pp, four seeds).

Table 6: Controller direction (distrust minus smooth) and Adam-NV minus Adam on CIFAR-100.

Comparison	Seeds	Δ (pp)	95% CI	Holm p
Direction, clean	4	+0.10	$[-0.23, +0.42]$	0.42
Direction, const. noise	4	+0.09	$[-0.07, +0.25]$	0.36
Direction, intermittent	4	+0.29	$[+0.01, +0.57]$	0.14
CIFAR-100, clean	8	−0.29	$[-0.62, +0.04]$	0.15
CIFAR-100, intermittent	4	−0.01	$[-0.64, +0.63]$	0.97

5.5 Computational cost

We timed Adam, Adam-NV, AdamW and AdaBelief on a Tesla T4 with a random input batch of 128 images in 20 rounds. In every round each optimizer is timed once, in a freshly randomized order, with 50 warm-up and 200 measured steps and GPU synchronization around each timed region; we report medians and ratios paired within a round, with bootstrap 95% intervals over rounds. The optimizer step takes 27.3 ms for Adam-NV against 3.8 ms for Adam (7.2 times as long, interval $[7.14, 7.18]$). The end-to-end training step (zero-grad, forward, backward, update) takes 126.5 ms against 108.0 ms, an increase of 17.1% (interval $[16.9, 17.5]$%); AdamW adds 0.4% and AdaBelief 2.8%. The end-to-end difference (18.5 ms) is smaller than the step-only difference (23.5 ms) because the forward and backward pass is 5.1 ms faster in the Adam-NV runs (99.0 against 104.2 ms). This persists under randomized order (Adam's total time varies by less than 1 ms across positions in a round), and we have not explained it. Adam-NV keeps four per-element state buffers (first moment, second moment, smoothed score and running product) against two for Adam, so its optimizer state, which counts only the buffers the optimizer keeps, is twice as large, 170.5 MiB against 85.3 MiB for our 11,173,962-parameter ResNet-18, and its peak allocated GPU memory, which also includes weights, gradients and activations, is 85.6 MiB higher (914.1 against 828.5 MiB, +10.3%). The benchmark script and its outputs are in the repository (the script `benchmark_optimizer_overhead_v2.py` in the `benchmarks` folder). The intervals describe round-to-round variation within one session on one GPU (the per-round ratio of Adam-NV to Adam ranges from 1.156 to 1.177), not variation across sessions or machines, and the figures apply to our unfused PyTorch implementation.

5.6 Does the score measure gradient noise?

We checked the proxy against direct measurements of gradient noise in two diagnostics on Adam-NV runs (clean labels and 20% label noise): a first with two seeds per condition and $K = 16$ minibatches, and a second, designed after the limits of the first were clear, with five seeds per condition, $K = 64$ and a rule fixed beforehand. The unit of replication is the seed, because the millions of elements within a run are not independent. After epochs 1, 5, 10, 20, 30, 45 and 60, at the fixed weights and without an optimizer step, we drew the minibatch gradients (BatchNorm in training mode) and computed for each element the direct noise fraction q (unbiased variance over mean square, unclamped), the absolute gradient scale, and, within each tensor, the split-half reliability of q, which is the ceiling for any correlation with it; we also recomputed the score with $\lambda = 0$ from the same optimizer state. Unit tests confirm that weights, BatchNorm statistics and optimizer state are unchanged, and with 20% label noise the runs reproduce the validation accuracy of the earlier Adam-NV runs with the same seeds at every epoch. The statistic is the median over the 20 convolution tensors of the within-tensor Spearman correlation of s with q, divided by the square root of the reliability of q.

The pre-registered outcome is "unresolved" in both conditions: at the primary checkpoints (epochs 30, 45 and 60) the mean reliability within convolution tensors is 0.04, below the required 0.2. It falls from about 0.7 at epoch 1 to 0.01–0.02 by epoch 45, and under label noise the mean q reaches 0.999, the value of pure noise, so there is nothing left to rank. Where the estimate is resolvable (epochs 5 and 10, reliability 0.44 and 0.41), the corrected within-tensor correlation is -0.02 (CI $[-0.03, -0.01]$) on clean labels and -0.01 (CI $[-0.02, -0.01]$) under label noise, and about $+0.01$ with $\lambda = 0$; this is exploratory, since the rule was fixed for epochs 30–60. Within convolution tensors the score instead depends on the absolute gradient scale: the correlation is $+0.22$ (CI $[+0.21, +0.24]$) on clean labels and $+0.22$ (CI $[+0.21, +0.23]$) under label noise at epochs 30–60, against -0.02 and -0.04 for the $\lambda = 0$ score, so the $\lambda|m|$ term of Section 3.1 is the source of that dependence. Between tensors the association is modest and not stable across conditions: the Spearman correlation of the 62 tensor means is $+0.26$ (CI $[+0.16, +0.35]$) on clean labels and -0.14 (CI $[-0.28, -0.01]$) under label noise at epochs 30–60. Pooled over elements, the values of the first diagnostic (0.79 at epoch 60 on clean labels, 0.02 under label noise) are reproduced (0.74 and 0.02). The diagnostics therefore do not validate the proxy: they show no within-tensor relation to measured gradient noise where that noise can be measured, a scale dependence that the $\lambda|m|$ term explains, and a between-tensor association that is not stable.

5.7 Memorization under label noise

Both Adam and Adam-NV reach about 87.5% accuracy at epoch ≈22 and then decline to roughly 78.8% as the corrupted labels are fitted. Held-out accuracy at the end is the same across all arms (78.4–78.9%), and the peak varies by less than 0.4 pp (87.3–87.7%). Memorization, however, differs (Table 9, Fig. 1). Adam-NV fits $83.5 \pm 0.5\%$ of the corrupted labels against $79.6 \pm 1.0\%$ for Adam, a difference of +3.91 pp (95% CI $[+3.11, +4.71]$), with Adam-NV higher in all eight seeds (exact sign-flip $p = 0.0078$, the minimum for eight seeds; Holm $p < 0.001$).

The constant-β_1 control matters here. Adam with a constant β_1 equal to the time average of Adam-NV's logged per-tensor mean (0.81) gains only +0.64 pp (CI $[-0.40, +1.67]$). Because Adam-NV's logged β_1 drifts from 0.87 at the first epoch to about 0.78 late in training, with the lowest per-tensor mean around 0.6 (Fig. 2; element-level values were logged only in the shuffle-control runs, Table 7), we also ran constant β_1 of 0.78 and 0.70. They gain +0.77 and +0.83 pp, neither significant, and Adam-NV stays 3.14 pp (CI $[+2.16, +4.13]$, Holm $p = 0.0005$) and 3.08 pp (CI $[+1.92, +4.24]$, Holm $p = 0.0012$) above them, in 8 of 8 seeds each. Over $0.9 \to 0.7$ the dose-response of memorization to constant β_1 is weak (at most 0.8 pp, none significant). Under the pre-specified rule the outcome is that the effect is not explained by any constant between 0.70 and 0.90.

Direction of the adaptation. The smooth response raises β_1 above 0.9, toward 0.999, when the score is high (logged per-tensor mean β_1 0.92, range 0.90–0.99). It gives the opposite result: it fits $78.11 \pm 0.89\%$ of the corrupted labels, 1.47 pp *fewer* than Adam (CI $[-2.40, -0.55]$, Holm $p = 0.007$, lower in 8 of 8 seeds), and 5.38 pp fewer than the distrust response (CI $[+4.55, +6.21]$, 8 of 8 seeds). Validation accuracy is again unchanged within noise (smooth minus Adam −0.42 pp, CI $[-0.95, +0.12]$). Under the rule fixed for that experiment the outcome was that the distrust direction drives the effect, but that rule did not include constants above 0.9, which we tested next. The experiment also does not show whether the element-wise link between β_1 and each element's own score is needed.

Constant β_1 above 0.9. The smooth response also raises the average β_1 (logged per-tensor mean 0.92), so we ran constants of 0.92 and 0.95 under a rule fixed beforehand. A constant β_1 of 0.95 fits $78.56 \pm 1.06\%$ of the corrupted labels, 1.03 pp fewer than Adam (CI $[-2.21, +0.16]$, lower in 6 of 8 seeds), which is statistically indistinguishable from the smooth response (smooth minus constant 0.95: −0.45 pp, CI $[-1.32, +0.43]$). A constant of 0.92 shows no reduction (+0.44 pp, CI $[-0.82, +1.71]$) and a larger spread across seeds (SD 1.79 against 0.9 to 1.3 for the other arms); the smooth response is 1.91 pp below it (CI $[-3.52, -0.30]$, Holm $p = 0.10$). The pre-specified rule gives "adaptation adds" for 0.92 and "reproduces" for 0.95, so the overall outcome is partial. Pooling all constant-β_1 runs from 0.70 to 0.95, memorization falls as β_1 rises (slope −5.7 pp per unit of β_1, $r = -0.38$, $p = 0.008$; descriptive only). We therefore read the smooth-response reduction as consistent with a higher average β_1 and not as a property of the adaptation, whereas the extra memorization of the distrust response is not mirrored by any constant we tested.

Which part of the controller matters. To separate the element-wise link from the distribution of β_1, we permuted Adam-NV's per-element β_1 across the elements of each parameter tensor before applying it, either with a fresh permutation every step or with one fixed permutation per tensor; within each tensor the distribution of values is unchanged, and their link to each element's own score is destroyed. A run through the same wrapper without permutation reproduced the two existing Adam-NV logs bit for bit (maximum difference 0 over all 15 logged columns). Both permuted variants keep most of the effect: they fit 2.40 pp (CI $[+1.20, +3.60]$) and 2.81 pp (CI $[+2.18, +3.44]$) more corrupted labels than Adam, 61% and 72% of Adam-NV's +3.91 pp, in 8 of 8 seeds each. They remain below Adam-NV by 1.51 pp (CI $[+0.93, +2.08]$, Holm $p = 0.002$) and 1.10 pp (CI $[+0.39, +1.81]$, Holm $p = 0.016$), and the two permutations do not differ detectably from each other. By the pre-specified rule the outcome is partial: most of the effect does not need the element-wise link, and a remainder of about 1.1 to 1.5 pp does. Validation accuracy is unchanged within noise (−0.20 pp for both variants, intervals including zero).

Tensor-level adaptation. Because a within-tensor permutation preserves each tensor's own mean, we also set every element of a tensor to that tensor's current mean β_1, which removes all within-tensor spread. A run of the same wrapper without change again reproduced the existing Adam-NV logs bit for bit, and it supplied the per-tensor statistics above. The tensor-mean control fits $82.13 \pm 0.70\%$ of the corrupted labels, 2.55 pp more than Adam (CI $[+1.56, +3.53]$, 8 of 8 seeds), which is 65% of Adam-NV's effect. It stays

1.36 pp below Adam-NV (CI $[+0.83, +1.90]$, Holm $p = 0.002$) and does not differ detectably from either permutation (tensor-mean minus shuffled-each-step +0.15 pp, minus shuffled-once −0.27 pp, both intervals including zero). By the pre-specified rule the outcome is partial: both levels contribute. The three controls agree, so the within-tensor spread of β_1 adds nothing we can detect.

The runs also give the element-level distribution of the controller's β_1 (Table 7). It is much closer to the base value than the per-tensor mean suggests: the element-weighted mean stays between 0.88 and 0.89, the median between 0.887 and 0.898, and at epoch 60 only 5% of the elements lie below 0.845 and 1% below 0.818; fewer than 0.07% fall below 0.7. The per-tensor mean of 0.78 in Fig. 2 is lower than the element-weighted mean because 41 of the 62 tensors are one-dimensional (normalization parameters and a bias, 9,610 elements in total) and have lower coefficients than the convolution tensors (Table 8: mean 0.75 against 0.85 at epoch 60); the 20 convolution tensors, which hold 99.9% of the elements, also fall below 0.9, with a lowest tensor mean of 0.64. The constant controls therefore lowered β_1 for every element more than Adam-NV does for most elements, which makes a matched constant a conservative comparison; because we tested constants from 0.70 to 0.95, which bracket the element-level mean, this does not change the conclusion that no constant we tested reproduces the effect.

Where the controller acts. The 41 one-dimensional tensors have the lowest β_1 on average, while the 20 convolution tensors hold 99.9% of the elements (Table 8). We therefore enabled the controller on only a subset of the parameters, with plain Adam ($\beta_1 = 0.9$) on the rest (Section 3.2): on the 41 tensors of rank at most one (normalization scales and shifts and the classifier bias; 9,610 elements, 0.09% of the parameters) or on the 21 tensors of rank at least two (20 convolutions and the classifier weight). Seeds and settings are those of Table 9, and the rule and its Holm family were written in a run book beforehand. Only on the one-dimensional tensors, the controller fits 1.19 pp more corrupted labels than Adam (CI $[+0.50, +1.88]$, higher in 7 of 8 seeds, Holm $p = 0.009$), 30% of Adam-NV's +3.91 pp; only on the higher-rank tensors it fits 2.73 pp more (CI $[+1.97, +3.49]$, 8 of 8, Holm $p = 0.0002$), 70%. Adam-NV stays above both, by 2.72 pp (CI $[+2.24, +3.19]$, Holm $p < 0.001$) and 1.18 pp (CI $[+0.50, +1.86]$, Holm $p = 0.009$), in 8 of 8 seeds each. The rule required one subset to reach 75% of the effect; neither does, so the outcome is that both subsets contribute. Validation accuracy is unchanged within noise (−0.16 and −0.20 pp, intervals including zero).

Three points are exploratory and not part of the rule. The two effects sum to 3.92 pp, close to +3.91 pp, and enabling the controller in one subset adds the same whether or not it is on in the other (+1.19 against +1.18 pp; +2.73 against +2.72 pp); the interaction contrast is +0.01 pp (CI $[-0.84, +0.87]$), which is compatible with additivity but cannot exclude about 0.9 pp (22% of +3.91 pp). The one-dimensional contribution is the least certain: it is not significant if Holm is applied over all five contrasts and four metrics (adjusted $p = 0.075$, as for Adam-NV minus the higher-rank-only arm; at most 0.0012 for the other three). Finally, tensors are grouped by rank and not by function. A constant β_1 of 0.79 (the time average of the controller's own value on these tensors) applied to the one-dimensional tensors only, with plain Adam elsewhere, fits 0.37 pp more corrupted labels than Adam (CI $[-0.90, +1.64]$, higher in 4 of 8 seeds), 31% of the adaptive effect. The adaptive arm exceeds this constant by 0.82 pp (CI $[-0.02, +1.66]$, 7 of 8 seeds, Holm $p = 0.11$; the rule was fixed beforehand), so by the rule the outcome is inconclusive: the data lean toward the adaptivity mattering on these tensors but do not establish it. Using the null run instead of Adam as the baseline changes the two subset contrasts little (+1.34 pp, CI $[+0.71, +1.97]$, and +2.88 pp, CI $[+2.07, +3.68]$).

Table 7: Element-level distribution of Adam-NV's β_1 under label noise (descriptive statistics from two runs, seeds 0 and 1; the 11.2 million elements within a run are not independent observations).

Epoch	Mean	Median	5% quantile	1% quantile	Minimum	Below 0.7 (%)
1	0.891	0.894	0.876	0.857	0.601	0.01
20	0.893	0.898	0.871	0.846	0.486	0.03
40	0.885	0.891	0.850	0.824	0.406	0.02
60	0.881	0.887	0.845	0.818	0.436	0.02

Other noise rates. With 10% and 40% symmetric label noise under the same protocol (Adam and Adam-NV only, six seeds, rule fixed beforehand), Adam-NV again fits more corrupted labels than Adam: +1.41 pp

Table 8: Adam-NV's β_1 by tensor type under label noise: mean over tensors (equal weight per tensor) at four epochs, from two runs. The last two rows give the mean over all tensors (each tensor counts once) and over all elements (each element counts once).

Tensor type	Tensors	Elements	Epoch 1	Epoch 20	Epoch 40	Epoch 60
One-dimensional	41	9,610	0.867	0.810	0.763	0.753
Two-dimensional	1	5,120	0.717	0.703	0.749	0.781
Convolution (4-D)	20	11,159,232	0.874	0.863	0.849	0.846
All tensors, equal weight	62	11,173,962	0.867	0.825	0.790	0.784
All elements (element-weighted)	62	11,173,962	0.891	0.893	0.885	0.881

Table 9: Label noise (20%), learning rate 10^{-3}, eight seeds. Memorization is the last-five-epoch mean fraction of corrupted training examples predicted as their wrong label; "Mem." is memorization; Δ is the paired difference from Adam in percentage points; "Seeds > Adam" counts seeds with higher memorization than Adam. "Rank ≤ 1" and "rank ≥ 2" restrict the controller to those parameter tensors. Contrasts are paired by seed.

Arm	Mem. (%)	Val. acc. (%)	Clean acc. (%)	Δ (pp)	95% CI	Seeds > Adam
Adam, $\beta_1 = 0.90$	79.58 ± 1.01	78.79	98.73	—	—	—
Constant $\beta_1 = 0.81$	80.22 ± 0.99	78.80	98.74	$+0.64$	$[-0.40, +1.67]$	7/8
Constant $\beta_1 = 0.78$	80.35 ± 1.11	78.85	98.75	$+0.77$	$[-0.45, +1.99]$	6/8
Constant $\beta_1 = 0.70$	80.41 ± 1.30	78.53	98.77	$+0.83$	$[-0.57, +2.22]$	6/8
Constant $\beta_1 = 0.92$	80.03 ± 1.79	78.58	98.74	$+0.44$	$[-0.82, +1.71]$	5/8
Constant $\beta_1 = 0.95$	78.56 ± 1.06	78.61	98.70	-1.03	$[-2.21, +0.16]$	2/8
Adam-NV	$\mathbf{83.49 \pm 0.54}$	78.84	98.84	$\mathbf{+3.91}$	$[+3.11, +4.71]$	8/8
Adam-NV, smooth response	78.11 ± 0.89	78.37	98.72	-1.47	$[-2.40, -0.55]$	0/8
Adam-NV, β_1 shuffled each step	81.98 ± 0.69	78.59	98.80	$+2.40$	$[+1.20, +3.60]$	8/8
Adam-NV, β_1 shuffled once (fixed)	82.40 ± 0.73	78.59	98.78	$+2.81$	$[+2.18, +3.44]$	8/8
Adam-NV, β_1 set to tensor mean	82.13 ± 0.70	78.61	98.83	$+2.55$	$[+1.56, +3.53]$	8/8
Adam-NV, rank ≤ 1 tensors only	80.78 ± 0.52	78.63	98.78	$+1.19$	$[+0.50, +1.88]$	7/8
Adam-NV, rank ≥ 2 tensors only	82.31 ± 0.83	78.59	98.82	$+2.73$	$[+1.97, +3.49]$	8/8

Adam-NV minus constant β_1: +3.27 pp (0.81), +3.14 pp (0.78), +3.08 pp (0.70); all CIs exclude 0 (Holm $p \leq 0.0012$; Holm families are the three contrasts of the control experiment and the four of the dose-response experiment). Adam-NV minus smooth response: +5.38 pp ($[+4.55, +6.21]$, Holm $p < 0.001$ within the two direction contrasts). Adam-NV minus shuffled-each-step: +1.51 pp ($[+0.93, +2.08]$, Holm $p = 0.002$); minus shuffled-once: +1.10 pp ($[+0.39, +1.81]$, Holm $p = 0.016$); shuffled-once minus shuffled-each-step: +0.41 pp ($[-0.53, +1.35]$); Holm over the five shuffle contrasts. Adam-NV minus tensor-mean: +1.36 pp ($[+0.83, +1.90]$, Holm $p = 0.002$); tensor-mean minus shuffled-each-step +0.15 pp ($[-0.65, +0.94]$) and minus shuffled-once −0.27 pp ($[-1.10, +0.57]$); Holm over four contrasts. Subset arms against Adam: Holm $p = 0.009$ (rank ≤ 1 only) and 0.0002 (rank ≥ 2 only); Adam-NV minus rank-≤ 1-only +2.72 pp ($[+2.24, +3.19]$, $p < 0.001$) and minus rank-≥ 2-only +1.18 pp ($[+0.50, +1.86]$, $p = 0.009$); Holm over these four contrasts.

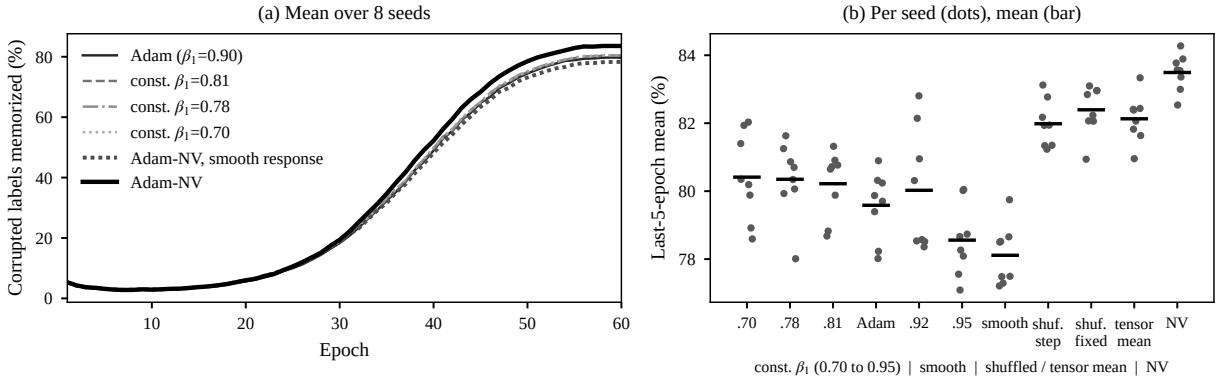


Figure 1: Memorization of corrupted labels under 20% label noise. (a) Mean over eight seeds per epoch. (b) Last-five-epoch value per seed (dots) and mean (bar); Adam-NV is separated from every constant-β_1 control, the smooth response lies below Adam, and the two shuffled variants and the tensor-mean control lie between Adam and Adam-NV.

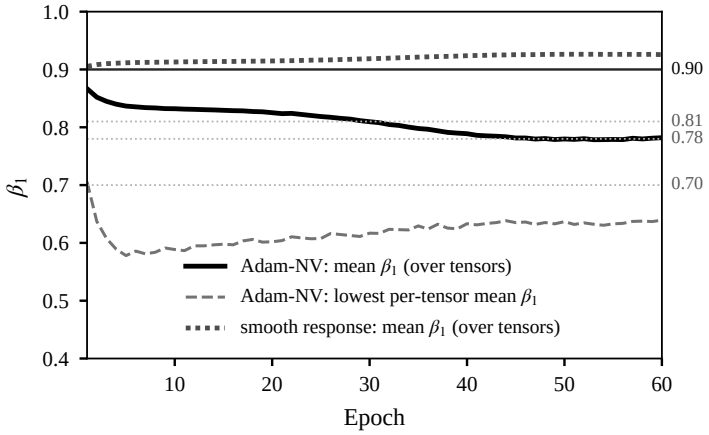


Figure 2: Adam-NV's momentum coefficient under label noise: mean over parameter tensors (each tensor's mean over its elements) and seeds, and the mean lowest per-tensor value, and, for the smooth response, the mean over tensors, against the constant-β_1 controls. Element-level values are in Table 7.

at 10% (CI $[+0.38, +2.44]$, higher in 6 of 6 seeds, Holm $p = 0.017$) and +2.66 pp at 40% (CI $[+1.78, +3.53]$, 6 of 6, Holm $p = 0.001$), against +3.91 pp at 20%. Validation accuracy is unchanged within noise (+0.11 pp, CI $[-0.21, +0.43]$, and −0.28 pp, CI $[-1.21, +0.65]$). The 10% result is the weaker one: its interval lies close to zero, one seed shows no difference, and with six seeds the exact sign-flip test cannot give an adjusted p below 0.0625. We ran no constant-β_1 or other controls at these rates, and we do not interpret the differences between rates, because the baseline memorization itself differs (91%, 80% and 64% for Adam).

5.8 Frozen tensor-wise momentum and memorization trajectories

We next asked whether the tensor-specific *average* first-moment coefficients alone could retain Adam-NV's memorization behavior, without step-wise adaptation or element-specific coefficient changes. From a separate Adam-NV pilot (seed 100) we froze 62 per-tensor coefficients and then compared the frozen optimizer against Adam-NV over five new, paired evaluation seeds (0–4); see Section 3.2. This is an additional control, not part of the original eight-seed Adam comparison. All results in this subsection are exploratory, and the reported confidence intervals are unadjusted.

At epoch 60, Adam-NV memorized 83.59% of deliberately corrupted training labels and the frozen control memorized 82.13%. The paired frozen-minus-Adam-NV difference was −1.45 percentage points (95% paired t CI $[-3.52, +0.62]$ pp; two-sided paired t $p = 0.123$; five pairs). Held-out test accuracy was 78.74% versus 78.49%, respectively (difference −0.25 pp, 95% CI $[-0.84, +0.34]$ pp; $p = 0.307$). The memorization difference favored Adam-NV in four of five evaluation seeds, but neither contrast establishes a nonzero population effect. More importantly, a confidence interval including zero is *not* a demonstration of equivalence. Individual-seed epoch-60 results appear in Appendix A.

Table 10: Experiment 22, ResNet-18/CIFAR-10 with 20% symmetric label corruption, five paired evaluation seeds (0–4). Values are at epoch 60, not last-five-epoch means. Adam-NV and frozen tensor-wise values are in %; "Memorization" is corrupted-label memorization and "Test accuracy" is held-out test accuracy. Differences (pp) are frozen minus Adam-NV, with unadjusted 95% paired t confidence intervals.

Measure	Adam-NV	Frozen	Diff. [95% CI]
Memorization	83.59	82.13	-1.45 $[-3.52, +0.62]$
Test accuracy	78.74	78.49	-0.25 $[-0.84, +0.34]$

The trajectory comparison (Fig. 3) shows that the frozen control closely tracks Adam-NV at selected memorization checkpoints: at epochs 20, 30, 45 and 60, Adam-NV records 6.00%, 19.57%, 68.82% and 83.59%, versus 5.80%, 19.35%, 67.54% and 82.13% for frozen momentum. Both exhibit rapidly increasing memorization after epoch 30 alongside falling held-out test accuracy. This joint timing is descriptive and does not establish that memorization caused the decline in test accuracy.

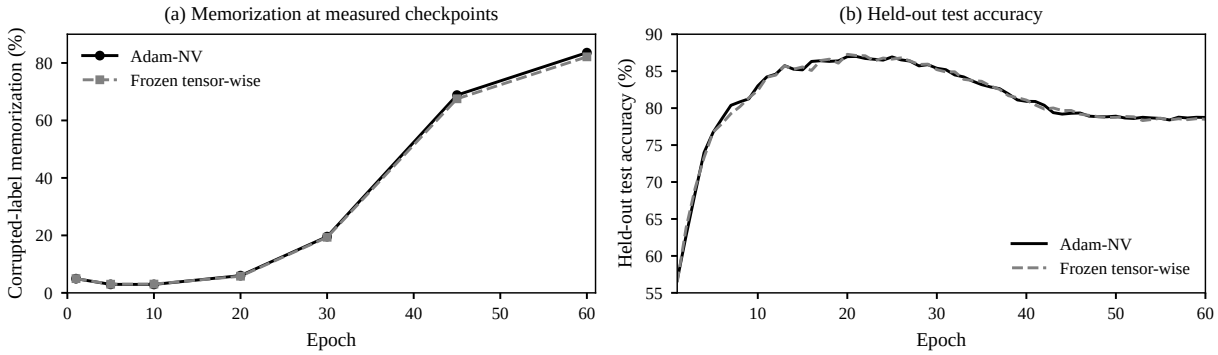


Figure 3: Experiments 22–23: (a) mean corrupted-label memorization at the seven measured checkpoints and (b) mean held-out test accuracy across 60 epochs, for Adam-NV and a frozen tensor-wise control over the same five paired seeds. Lines connect observed checkpoint means and are descriptive; no per-checkpoint hypothesis tests are inferred.

A fixed tensor-level profile therefore retains much of the memorization associated with Adam-NV in this tested setting, consistent with the earlier tensor-mean and within-tensor shuffling controls. However, the frozen coefficients were estimated from a separate pilot and can differ from the coefficients that the five Adam-NV runs would have used at each step. This experiment removes both element-level variation and temporal variation, so it does not isolate their individual causal contributions. The five-seed uncertainty also permits a practically relevant difference between methods; we neither conclude equivalence nor claim that temporal adaptation has no effect.

5.9 Second architecture: accuracy and memorization

To test whether the findings depend on the ResNet, we repeated the clean and label-noise comparisons on a small Vision Transformer (patch size 4, width 192, depth 6, 3 heads, pre-LayerNorm, no BatchNorm, 1.8 million parameters, no pretrained weights) at a learning rate of 5×10^{-4}, chosen beforehand and not tuned for any optimizer, with five seeds and the other settings of Section 4. The first arms are Adam, Adam-NV and Adam with a constant β_1 of 0.85, a value fixed in the run book before the runs; the rule and the Holm family (three contrasts per setting) were also fixed beforehand. All runs exceed 50% validation accuracy at epoch 5, and the models reach about 80% on clean labels. A five-epoch pilot on clean labels (seed 0) at this learning rate, run before the main runs as a go/no-go check, trained all three optimizers to 0.63–0.64 validation accuracy at epoch 5 against the threshold of 0.35 fixed in the run book; its logs were not retained, and its first-epoch accuracies equal those of the main runs. On clean labels Adam-NV reaches $80.27 \pm 0.47\%$ against $79.79 \pm 0.69\%$ for Adam: +0.48 pp (CI $[-0.06, +1.03]$, higher in 4 of 5 seeds, Holm $p = 0.14$), which by the rule is no detectable difference; the interval does not exclude a gain of about 1 pp, and five seeds can detect about 0.7 pp at 80% power. The constant of 0.85 gives +0.02 pp (CI $[-0.88, +0.92]$), and Adam-NV minus the constant is +0.46 pp (CI $[+0.07, +0.86]$, 5 of 5 seeds, Holm $p = 0.09$). With 20% label noise, Adam-NV fits $56.03 \pm 4.57\%$ of the corrupted labels against $52.18 \pm 5.15\%$ for Adam: +3.85 pp (CI $[+2.72, +4.98]$, 5 of 5 seeds, Holm $p = 0.002$), almost the size of the ResNet result (+3.91 pp), with validation accuracy unchanged (−0.25 pp, CI $[-1.73, +1.23]$).

The constant of 0.85 turned out not to be matched to the β_1 that Adam-NV logged on the ViT (0.79 on clean labels and 0.80 under label noise, dominated by the 52 one-dimensional tensors among its 80). We therefore added constants of 0.70, 0.80, 0.92 and 0.95 (within the 0.70–0.95 range of the ResNet controls)

under a rule written beforehand, with the same thresholds as for the ResNet controls and the matched 0.80 deciding the outcome (Table 11). Memorization falls steadily as the constant rises. The matched constant fits +4.83 pp more corrupted labels than Adam (CI $[+1.91, +7.75]$, 5 of 5 seeds, Holm $p = 0.050$), 1.25 times the Adam-NV effect (seed-bootstrap interval 0.75 to 1.76), and Adam-NV minus this constant is −0.98 pp (CI $[-3.91, +1.95]$); by the rule the outcome is that the effect is explained by the constant; this is a failure to distinguish the two, not a demonstration that they are equivalent. Adam-NV lies between the constants 0.80 and 0.85, at the level of a constant of about 0.82. Within every seed memorization decreases as β_1 rises (exploratory, one slope per seed: −39 pp per unit of β_1, CI $[-74, -5]$). The constants 0.70 and 0.85 give +5.93 pp (CI $[-1.98, +13.84]$) and +2.54 pp; 0.92 and 0.95 give −0.30 and −4.84 pp. For 0.92 and 0.95 the rule returns "adaptation beyond the constant", but these constants are higher than Adam-NV's own β_1, so the comparison is not matched and does not bear on adaptation. Validation accuracy is unchanged by every constant (−0.36 to −0.69 pp, intervals including zero). Adam-NV also fits the correctly labelled training examples better (+0.73 pp, CI $[+0.59, +0.87]$; exploratory), as do the lower constants. Two limits matter: the matched constant's Holm-adjusted p is 0.050 and the interval for 0.70 spans −2 to +14 pp, and memorization was still rising at epoch 60 in every arm.

Table 11: ViT-tiny, 20% label noise, five seeds: memorization (last-five-epoch mean) by constant β_1. Δ is the paired difference from Adam. Holm over six contrasts (0.70, 0.80 and 0.95 against Adam: $p = 0.32$, 0.050, 0.21; Adam-NV minus each: 0.81, 0.81, 0.034); the other constants are outside the family. The 0.85 arm was fixed before the runs; the others were added after it proved not to be matched to Adam-NV's logged β_1 (0.80).

Arm	Memorization (%)	Δ vs Adam (pp) [95% CI]	Above Adam	Adam-NV minus arm (pp) [95% CI]
Constant $\beta_1 = 0.70$	58.11 ± 5.14	$+5.93$ $[-1.98, +13.84]$	4/5	-2.08 $[-9.88, +5.71]$
Constant $\beta_1 = 0.80$ (matched)	57.01 ± 4.03	$+4.83$ $[+1.91, +7.75]$	5/5	-0.98 $[-3.91, +1.95]$
Constant $\beta_1 = 0.85$	54.71 ± 5.27	$+2.54$ $[+0.31, +4.77]$	5/5	$+1.31$ $[-1.71, +4.34]$
Adam, $\beta_1 = 0.90$	52.18 ± 5.15	—	—	$+3.85$ $[+2.72, +4.98]$
Constant $\beta_1 = 0.92$	51.88 ± 5.23	-0.30 $[-2.89, +2.30]$	1/5	$+4.15$ $[+1.43, +6.86]$
Constant $\beta_1 = 0.95$	47.34 ± 3.26	-4.84 $[-9.76, +0.08]$	1/5	$+8.69$ $[+4.23, +13.14]$
Adam-NV	56.03 ± 4.57	$+3.85$ $[+2.72, +4.98]$	5/5	—

6 Discussion

Two contributions, two strengths. The two parts of this study differ in strength. The accuracy result is negative and carefully controlled: we find no detectable benefit of Adam-NV, and its diagnostics do not support the noise proxy, so we do not claim that it is an effective noise-adaptive optimizer. The memorization result is the clearest effect of the adaptation that we found; it is an effect on training behavior, not an accuracy gain, and it concerns one adaptive-momentum method, not adaptive momentum in general.

Accuracy: what the evidence supports. Across clean training, three learning rates, three noise schedules and two datasets, Adam-NV does not exceed Adam by an amount that survives correction for multiple comparisons: the exploratory four-seed gain at learning rate 3×10^{-3} (Holm $p = 0.014$) is not confirmed on ten seeds (Holm $p = 0.057$). The only candidate, +0.17 pp at 3×10^{-3}, is not significant after multiplicity correction on ten seeds and is only partly reproduced by constant-β_1 Adam (+0.06 pp), so we cannot tell whether the remainder reflects the adaptation or chance; its upper confidence limit is about 0.3 pp. Because the constant controls recovered only about a third of the point estimate, we do not claim that a lower fixed β_1 is a substitute, and we did not run a budget-matched tuning study. Against our re-implementation of the other β_1-adaptive method, Adam-NV is better at 10^{-3} and 3×10^{-3} and tied at 3×10^{-4}, so Adam-NV's moment-based score gave a better adaptation than gradient direction in our setting, although neither beats a fixed β_1; given the diagnostic below, we do not read this as evidence that the score measures noise.

Memorization under label noise: what the results do and do not show. This is the clearest effect of the adaptation in our study. On ResNet-18 the extra memorization is not reproduced by a single global constant β_1 in the tested range, and the smooth-response ablation shows that the direction of the

adaptation matters: raising β_1 where the score is high lowers memorization below Adam's level. A constant β_1 of 0.95 reproduces that reduction within noise, so the reduction is not clearly specific to adaptation, whereas the increase under the distrust response is. The shuffle and tensor-mean controls narrow down the cause. Permuting β_1 within each tensor, every step or once, and replacing it by the tensor's mean each keep about 60 to 70% of the extra memorization (65% for the tensor mean), and the three do not differ detectably. The within-tensor variation therefore has no detected additional effect in these controls. The new pilot-calibrated, static per-tensor control (Table 10) also closely tracks Adam-NV over five evaluation seeds: at epoch 60 frozen momentum memorizes 82.13% versus 83.59% for Adam-NV (frozen minus Adam-NV -1.45 pp; 95% CI $[-3.52, +0.62]$). This further suggests that static differences across tensors can carry a substantial portion of the effect, but the interval does not establish equivalence, and the design does not distinguish all contributions of time versus element-level variation. Approximately 2.4 to 2.8 pp of the increase persists when only tensor-level coefficients are retained; these results do not distinguish time variation from a tensor-specific fixed coefficient. The remaining 1.1 to 1.5 pp is not recovered by those controls, but cannot uniquely be assigned to per-element score matching. Restricting the controller to subsets of tensors shows that the tensor-level part is not in one place: enabling it only on the 41 one-dimensional tensors (0.09% of the parameters) gives about 1.2 pp, and only on the other 21 tensors about 2.7 pp, which roughly add up to the full effect (the interaction interval, about ± 0.9 pp, is too wide to call them additive). A constant β_1 of 0.79 on the one-dimensional tensors only gives about a third of that effect ($+0.37$ against $+1.19$ pp), and the difference of 0.82 pp (CI $[-0.02, +1.66]$) is suggestive but not established, so we cannot say whether the adaptivity, rather than a lower β_1 on those tensors, raises memorization on ResNet-18. The effect did not change held-out accuracy at the end of training, so we do not claim it is harmful for generalization here, and an early-stopped model would not show a difference.

A second architecture. On a small ViT, Adam-NV again shows no detectable clean-accuracy gain, and the label-noise difference reappears at about the same size ($+3.85$ against $+3.91$ pp), but here a constant β_1 matched to Adam-NV's produces a comparable increase that the experiment does not distinguish from it, and memorization falls steadily as β_1 rises. The ViT effect is therefore consistent with what a lower β_1 produces, and we have no evidence that it reflects noise-awareness. The ResNet result, where no single global constant did, is specific to that architecture in our data. We have no explanation for the difference: the ViT has LayerNorm and no BatchNorm, among other differences, and we did not test which of them matters.

What the proxy diagnostic shows. The extra memorization cannot be attributed to noise-awareness, because the diagnostics do not support the score as a measure of per-element gradient noise: where the direct estimate is resolvable the within-tensor rank correlation is about zero, even without the $\lambda|m|$ term, and later in training the estimate cannot rank elements within a tensor at all, so the proxy cannot be validated there either. What the score does depend on within tensors is the absolute gradient scale. Together with the tensor-level results above, this is compatible with a controller that acts on tensors, but we did not test whether tensor-level scores reflect noise or gradient scale, and a design that resolves noise late in training would need many more minibatches per element or a different estimator.

Threats to validity. (i) All results use CIFAR-scale data, with ResNet-18 throughout and one small ViT with five seeds, one learning rate and one constant control; optimizer rankings can change with scale. (ii) Hyperparameters were not tuned for any optimizer, including Adam-NV; a tuned Adam-NV could differ. (iii) Several experiments use four seeds, so confidence intervals are wide (half-widths of 0.2 to 0.3 pp, against about 0.1 pp for ten seeds) and the sign-flip test cannot go below 0.125; "no detectable difference" is not equivalence. (iv) The CIFAR test set doubles as validation, although no model selection uses it; this affects the accuracy results, whereas the memorization measure is computed on the training set. (v) Label noise is symmetric and uniform and was tested at 10%, 20% and 40% with one learning rate, with controls only at 20%. (vi) Holm families were defined by us per experiment and are stated in the table notes. (vii) We re-implemented GONO; its ϵ placement differs from the authors' released code in a way we measured on synthetic gradients but did not evaluate on CIFAR. (viii) Timing was measured on one GPU type. (ix) The logged β_1 statistics are per-tensor means; the element-level and per-tensor-type statistics come from only two label-noise runs, and the matched constants were set from the per-tensor statistic. (x) The smooth response also raises the average β_1, and a constant β_1 of 0.95 reproduces its reduction within noise, so that reduction is not clearly specific to adaptation; the 0.92 arm also had an unusually large spread across seeds (SD 1.79).

(xi) AdaBelief and GONO were run with literature-default settings and no tuning (AdaBelief with $\epsilon = 10^{-8}$, no weight decay, no rectification); AdaBelief is known to be sensitive to ϵ, so its lower accuracy here is not a general ranking. (xii) The tensor-subset groups are defined by rank, not function (normalization scales, shifts and a bias; convolutions and the classifier); the constant-β_1 control on the one-dimensional tensors was inconclusive (adaptive minus constant +0.82 pp, CI −0.02 to +1.66), so adaptivity is not cleanly separated from a lower β_1; the plain tensors pass through the Adam-NV code, and the whole-model $\gamma = 0$ null run was inconclusive at the registered ±0.5 pp margin (interval −0.88 to +0.58 pp, paired SD 0.88 pp); the additivity interval is wide; and the one-dimensional contribution does not survive Holm over all contrasts and metrics. (xiii) The proxy diagnostic has five seeds per condition in its second version and two in its first; correlations are over dependent elements and are summarized per seed; the direct noise estimate cannot rank elements within a tensor late in training (reliability 0.04 with $K = 64$), so the pre-registered outcome is unresolved and the evidence for no tracking rests on epochs 5 and 10, outside the registered checkpoints; the per-tensor reliability correction assumes a measurement-error model that we did not test beyond synthetic controls; and only 0% and 20% noise were measured. (xiv) The new frozen tensor-wise study uses five seeds, a pilot-calibrated static coefficient map and only fixed-checkpoint memorization measurements; its epoch-60 metric differs from the last-five-epoch averages in earlier experiments, and frozen minus Adam-NV memorization has a wide CI $[-3.52, +0.62]$ pp. These results do not establish equivalence or isolate temporal and within-tensor effects separately. (xv) In the ViT experiment the learning rate (5×10^{-4}, no warmup) was not tuned and the model reaches only about 80% accuracy; five seeds give wide intervals (about ±0.5 pp for clean accuracy, and −2 to +14 pp for the 0.70 constant) and an exact sign-flip test that cannot go below 0.0625; the matched constant's Holm-adjusted p is 0.050; the constants 0.70, 0.80, 0.92 and 0.95 were added after the 0.85 control proved not to be matched, although their rule was written beforehand; and memorization was still rising at epoch 60.

7 Conclusion

We studied Adam-NV for two purposes. As an optimizer it gives no detectable accuracy improvement over Adam in the tested settings: its one candidate gain (at most about 0.3 pp at 3×10^{-3}, not significant after correction) is only partly reproduced by a lower constant β_1, it costs about 17% more per step, and a direct diagnostic with five seeds per condition does not validate its score: the registered test is unresolved, and an exploratory analysis at early checkpoints finds no within-tensor relation between the score and measured gradient noise. We therefore do not claim that Adam-NV is an effective noise-adaptive optimizer.

What the adaptation does change is memorization under label noise, which is our strongest finding. Under 20% label noise Adam-NV memorizes more corrupted labels than Adam (3.9 pp on ResNet-18 in all eight seeds, and also at 10% and 40% noise and on a small ViT) without changing held-out accuracy. On ResNet-18 no single global constant β_1 from 0.70 to 0.95 reproduces the effect, and substantial portions persist after removing within-tensor coefficient specificity: replacing β_1 by its tensor mean, or permuting it within tensors, keeps about two thirds of it, and enabling the controller on only the 41 one-dimensional tensors, or on only the other 21, reproduces about 30% and 70% of it, so neither subset alone suffices. In an additional five-seed ResNet study, an independently calibrated frozen per-tensor profile follows similar memorization trajectories and produces an epoch-60 mean of 82.13% compared with 83.59% for Adam-NV; the CI for their difference includes both zero and meaningful effects, so this does not prove equivalence. On the ViT a constant β_1 matched to Adam-NV's produces a comparable increase that the experiment does not distinguish from it, so there the effect is consistent with what a lower β_1 produces; the ResNet result that no constant does so is specific to that architecture in our data. These findings concern one adaptive-momentum method. For work on adaptive hyperparameters, they suggest reporting the adapted quantity's trajectory, comparing against a constant control matched to it, and, under label noise, measuring memorization directly. Code, logs, run books, decision rules, and scripts for recomputing reported results are retained in the research archive; an anonymized supplementary package is being prepared for review.

A Experiment 22: Seed-Level Endpoint Results

For transparency, Table 12 shows epoch-60 outcomes in percent for the independent five-seed evaluation set. Each matched pair uses the same initialization and label-noise seed, while the tensor-wise coefficient map was estimated using a distinct pilot seed (100) and fixed prior to evaluation. Memorization was collected at seven specified checkpoints (1, 5, 10, 20, 30, 45 and 60); held-out test accuracy was recorded each epoch.

Table 12: Per-seed epoch-60 results (%). Memorization (mem.) is corrupted-label memorization; test is held-out test accuracy. Δ is frozen minus Adam-NV (pp).

Seed	NV mem.	Frozen mem.	Δ mem.	NV test	Frozen test	Δ test
0	84.47	81.59	−2.88	78.65	78.98	+0.33
1	83.77	81.59	−2.18	78.87	78.80	−0.07
2	82.66	83.39	+0.73	78.49	78.46	−0.03
3	83.87	81.03	−2.84	78.57	77.76	−0.81
4	83.16	83.07	−0.09	79.11	78.45	−0.66

Reproduction and reporting limits. The corresponding training log archive includes ten evaluation CSVs, the independent pilot coefficient map, run configurations, success records, and analysis tables. The pilot map has 62 tensor keys and is shared by all five frozen runs. We do not claim that five pairs establish equality of the two optimizers, and do not count individual epochs or images as independent replications. The checkpoint-only training memorization measure must not be confused with last-five-epoch averages used in earlier experiments.

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